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Day Month Year

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Status of Thesis

Note: red = pre first edit, orange = pre second edit and beyond, green = done, blue = may not make it into the thesis, or it may be moved to the appendix.

- Chapter ??: Not started. Will write last/almost last
- Chapter ??: Apparently this is a good chapter...
- Chapter ??: Passed first drafting. Very Minor corrections to be made.
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's edits (see email from 29/09/2020 'quick question' for notes).
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's comments.
- Chapter ??: Passed first drafting. Still to apply Ken and bryan comments
- Chapter ??: Not written, vaguely planned
- Chapter 1: Significant amounts of thought has already gone into this.
- Chapter ??: Not started. Will write last/almost last

Chapter 1

Characterisation of InGaAs

Photodiodes for detecting $2\mu\text{m}$ Light

1.1 Good references

- 95% QE diode with cut-off at $2.4\mu\text{m}$ [1].
- semiconductor book [2]
- photodiode/optics book [3]

1.2 Introduction

The current generation of gravitational will be limited by the thermal noise due to brownian motion of the molecules that make up the mirror coatings. Since these thermally driven fluctuations are a limiting source of noise [4], proposed future gravitational wave detectors such as the Einstein Telescope [5] and LIGO Cosmic

Explorer [6] will use cryogenically cooled mirrors.

However, cryogenically cooled mirror coatings will not be compatible with the current laser wavelength, which is $1\text{ }\mu\text{m}$, and so future gravitational wave detectors may use either a $1.5\text{ }\mu\text{m}$ or $2\text{ }\mu\text{m}$ wavelength laser. Currently, $2\text{ }\mu\text{m}$ is of interest as there have been promising designs for low noise mirror coatings that are optimised for this wavelength.

Currently, as InGaAs is widely used semiconductor for detecting $1\text{--}1.6\text{ }\mu\text{m}$ light and thus suitable InGaAs photodiodes were available for detecting the gravitational-wave-signal-containing laser beam that emerges from the OMC. These photodiodes are low noise, linear, can detect many tens of milliwatts of light, and have a high quantum efficiency. Additionally, InGaAs is a popular material to use as it can be operated at room temperature.

As the current InGaAs photodiodes are not sensitive to $2\text{ }\mu\text{m}$ light, ‘extended’ InGaAs photodiodes may be used instead. These diodes are extended in the sense that their cut-off wavelengths are above $2\text{ }\mu\text{m}$. To achieve this longer cut-off wavelength, extended InGaAs contains a higher percentage of indium than normal; however, this extra indium comes at the expense of the structural integrity of the lattice structure. This can cause the quality of the photodiode to be degraded. Previous experiments have been limited by the quantum efficiency of the extended InGaAs photodiode that was used [7,8]. To obtain low enough electronic noise, the authors of [7] used the photodiodes without a bias. A photodiode with a quantum efficiency of up to 95% has been reported.

Alternative materials that could be used include HgCdTe and InSb. HgCdTe has potential, however there are still technical challenges to overcome in creating diodes that work at???. The properties of an InSb detector with a high reported QE is reported in chapter ??, however this detector would not be suitable for use in a gravitational wave detector.

In this chapter, the relationship between quantum efficiency, reverse bias and noise in Extended InGaAs Photodiodes is explored. A selection of extended InGaAs photodiodes were tested with the aim of understanding general trends in the behaviour of extended InGaAs diodes. It was found that to get a high quantum efficiency and linearity, the diodes must be operated with a bias. However, if the diodes were used with a high bias then the dark noise of the diode would exceed the shot noise level of the detected light. The dark noise could be reduced significantly by cooling the diodes and so future detectors may feature cooled read-out diodes. Generally speaking, the longer the cut-off wavelength, the worse the diodes noise was. Future work may include investigating the growth conditions for extended InGaAs photodiodes as it has been shown that the dark noise of a diode made from extended InGaAs can be reduced by orders of magnitudes by optimising this. However, at the time of writing (2020), commercially available InGaAs photodiodes for detecting 2 μm light that are suitable for use in a gravitational wave detector do not exist.

1.3 What is an Extended InGaAs Photodiode

A photodiode is a square law detector and so it produces a current that is proportional to the power of the light it is detecting. Usually, they are of the order 1 mm in

diameter. They are two-terminal devices, with an anode and a cathode. Photodiodes often follow a p-doped, intrinsic, n-doped (PIN) structure. PIN photodiodes are used in applications where there is milliwatts of light power.

To understand a PIN photodiode, it is useful to first consider a simpler PN photodiode. A pn junction consists of two types of semiconductor: p-doped and n-doped. In the P-doped region, holes (the majority charge carriers) are free to move whereas in the n-doped region electrons (the majority charge carriers) are free to move.

At the interface between the p-doped and n-doped region, majority carriers from each side diffuse across it. Then, the minority charge carrier combines with a majority charge carrier, and so no net current will flow out of the device. When in equilibrium, there are no free carriers in the p and n regions, and so these regions become charged. Due to the diffusion of majority charge carriers across the interface, there is an electric field between the p-doped and n-doped region. The region of the device containing no free carriers is called the depletion region.

When a photon is incident on a pn junction, it will create an electron-hole pair. If this photon is absorbed within the depletion region of the diode, then due to the electric field, the electron and hole will be separated. These electrons/holes will generate a current in the circuit connected to the diode. However, if a photon is absorbed outside the depletion region, then the electron-hole pair will not be separated and will recombine. These photons will not contribute to the current in the circuit.

To widen the depletion region, an external voltage can be used. When *reverse bias* is applied to a photodiode – **that is a positive voltage across the cathode and anode** –, there is a larger electric field difference at the interface between the p-doped and n-doped region. This increases the volume of the photodiode results in absorbed photons contributing to the photocurrent.

To further increase the width of the depletion region, a layer of intrinsic semiconductor can be used to separate the p and n-doped region. An intrinsic semiconductor has no charge as it has an equal number of excited electrons and holes. An electric field will be present in a larger volume of material and so more photons will be converted into electrons which leave the photodiode diode.

InGaAs is a widely used semiconductor for making photodiodes. Typically, the indium content is around 53%, sometimes this is made clear by writing $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$, so that the InGaAs can be lattice matched to InP substrate on which the photodiode is grown on. At this level of indium doping, the InGaAs photodiode will be suffer from few mechanical defects and so the diodes created with it will be of high quality. Typically they are sensitive up to wavelengths around 1.6 μm .

Longer wavelength sensitivity can be achieved by increasing the amount of indium doping. These photodiodes are often called ‘extended’ and can be sensitive for wavelengths up to 2.6 μm . One of the main issues faced when creating an extended InGaAs photodiode is the lattice mismatch between the InGaAs and the InP substrate on which its grown. Lattice mismatching can arise due to a strained crystal structure, and since the extra indium in an extended InGaAs photodiode causes this [9] it is hard to fabricate a low noise extended InGaAs photodiode [10]. Above

53% indium, a buffer layer of InGaAs with less indium in between the absorbing layer and the substrate can be used to mitigate the lattice mismatch slightly, but the dislocations still give rise to trap states which cause the generation and recombination of charge carriers [11]. Lattice mismatching is the source of the large amounts of dark current found in extended InGaAs as well as the $1/f$ noise.

similar circuit model?

1.4 What is Needed from the Read-out Photodiode in a Gravitational Wave detector

The photodiodes used in the readout of a gravitational wave interferometer should have a quantum efficiency of more than 99%. The reason for having such a high quantum efficiency requirement is that to achieve the baseline level of ~ 10 dB of squeezed light that future gravitational wave detectors will use, losses in the interferometer have to be extremely low as losses degrade the amount of squeezing achieved.

Photodiodes for gravitational wave detectors should have low $1/f$ dark noise so that they are not limited in the detection band by electronic noise. Third generation ground based gravitational wave detectors will be quantum noise limited down to about 10 Hz [12], so the $1/f$ noise must be below the shot noise of the photo-current above this frequency.

The $1/f$ noise could be avoided by using a heterodyne detection, however the shot noise would increase if the mixing of the modulation and demodulation waveforms was not time independent [13] and this limits the choice of waveforms. These schemes recover increase the SNR by increasing the signal, however the use of squeezing means that this approach would still not be sufficient as the noise would still be enhanced compared to a homodyne readout. However, it has been proposed that the increase in noise associated with heterodyne readout could be avoided by the use of a broadband squeezer [14].

Current gravitational wave detectors operate in the 1-10 mW range at the output [15] to ensure that the junk light from other modes is a small portion of the total signal. Thus, any new photodiode should be able to produce ~ 10 mA of photocurrent.

The properties were investigated in a selection commercially available photodiodes. The photodiodes that were tested are listed in table ???. According to each of the diodes datasheets, none of them have 99% quantum efficiency; however, the quantum efficiency of a diode can be improved by increasing the bias voltage (see Section ??). Increasing the bias increases the dark current and dark noise of the photodiode and so for each diode a maximum reverse bias voltage was found. The bias voltage also affects the saturation limit of the photodiode. If a diode was found to have a low saturation threshold (~ 1 mA), then the diode is not fit for use in a gravitational wave detector.

1.5 The effect of reverse bias on an Extended In-GaAs Photodiode -FD10D

To demonstrate the effect that reverse bias has on the quantum efficiency of a diode, the Thorlabs FD10D was used. The FD10D was exposed to a variety of beams so that saturation effects would be visible. Two styles of measurements were used: One where the reverse bias and the response of the diode were measured at DC, and a lock-in style measurement where the quantity $\frac{\Delta\eta}{\Delta V_B}$ was measured. The quantum efficiency should be independent of frequency if the diode is operating in a linear regime, and so the two styles of measurements should yield the same outcome. To prove that the increase in signal is a result of increasing quantum efficiency and not some other mechanism, for instance an avalanche effect, the noise of the photocurrent was compared to the level expected for shot noise at a bias voltage that yielded an acceptable level of dark noise (see Section ?? for details.).

1.5.1 Apparatus

To test the dependence of the quantum efficiency of a photodiode as a function of reverse bias, a photodiode circuit with an externally supplied bias was used. This is shown in Figure 1.1. A DC bias was created by low-passed bench-top lab power supply. An AC component could be added onto the bias via a summing OP-amp. The FD10D was connected to a transimpedance amplifier that whose electronic noise was below the shot noise level for ~ 10 mA currents.

A 1550 nm wavelength laser (Pure Photonics PPL) was used to illuminate the photodiode (FD10D). The beam was split by a balanced beam splitter so that the

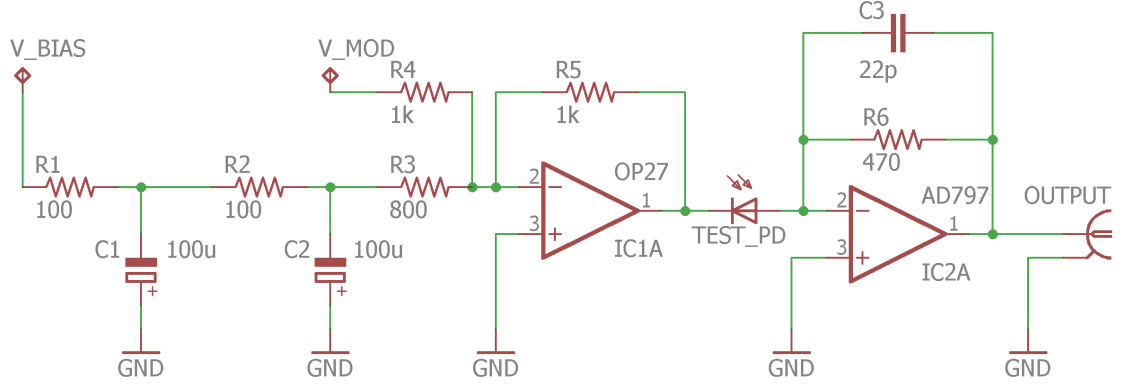


Figure 1.1

light could be amplitude stabilised. The in-loop diode was a high performance InGaAs photodiode (C3061GH), and the out-of-loop diode was the diode under test. The laser was amplitude stabilised by actuating on the laser diode's current. Shot noise limited measurements could be made above 1 kHz. A half waveplate and polarising beam splitter were used to control the power level of the light incident on the FD10D, and a lens was used to set the size of the beam.

1.5.2 Method

The FD10D was exposed to a variety of different beams to investigate the possibility that current density related phenomena could affect the results. This was done by changing the beam size and beam power. As the FD10D is 1 mm, the maximum beam width would be around 330 μm in order for less than 1% clipping losses. The powers used were in the milliwatt range as these types of beam powers are the ones that a gravitational wave detector would operated with. The methods in this section can only be used to measure relative increases in quantum efficiency, measurements of absolute quantum efficiency require [find ref ??](#) additional apparatus.

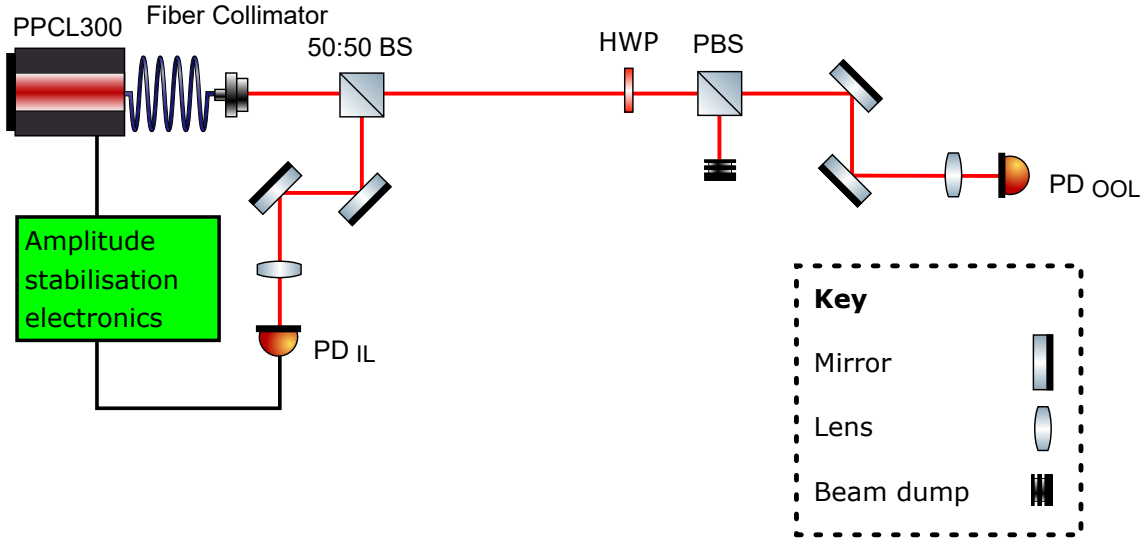


Figure 1.2

To measure the effect that changing the reverse bias voltage has on the quantum efficiency of a photodiode, two approaches were used. By measuring the increase in photocurrent at DC it is straight forward to determine the increase in quantum efficiency. Alternatively, one can perform a lock-in style measurement. If the quantum efficiency depends on the reverse bias of the photodiode, then modulating the reverse bias, V_B at a frequency f will result in η modulating at f , and so the measurement will be of $\Delta\eta/\Delta V_B$. The input modulation $\Delta V_{\text{in}} = \Delta V_B$ and the output modulation $\Delta V_{\text{out}} \cos \theta = \Delta\eta V_{\text{TIA}}(V_B)$, where $V_{\text{TIA}}(V_B)$ is the DC signal from the photodiode at a reverse bias V_B and θ is the phase between the input and output modulation.

As 0.6 V reverse bias was the highest bias that gave an acceptable level of dark noise (see section ??), the noise of the photocurrent was measured at a region where the noise of the laser was limited by shot noise. This was done to ensure that the

increase in photocurrent was due to a real increase in quantum efficiency and not due to another mechanism e.g. an avalanche process. The expected level of noise, n , of the photocurrent, in units of $A_{\text{rms}}/\sqrt{\text{Hz}}$, is

$$n = \sqrt{2qI_{\text{OOL}} + 2qI_{\text{IL}} + n_{\text{IL dark}}^2 + n_{\text{OOL dark}}^2 + n_{\text{loop}}^2 + n_{\text{analyser}}^2} \quad (1.1)$$

where q is the charge of the electron, $I_{\text{OOL/IL}}$ is the current through the out-of-loop and in-loop photodiodes, $n_{\text{OOL/IL dark}}$ is the dark noise of the out-of-loop diodes and in-loop photodiode, n_{loop} is the electronic noise of the loop and n_{analyser}^2 is the noise of the analyser. By tracking the height of an intensity modulation in the light for a series of powers up to 16 mW, the linearity of the diode could be investigated.

At this bias, the quantum efficiency was calibrated using two other power meters. The power incident onto the photodiode was measured and the power reflected by the window at small angles was measured.

1.5.3 Results

The effect of increasing the reverse bias on the quantum efficiency of the FD10D is shown in Figure 1.3. For all beams intensities that the photodiode was exposed to, the increase in quantum efficiency was $\sim 15\%$ as the reverse bias went from 0 V to 1.8 V, the stated maximum reverse bias in the datasheet of the FD10D. There may be a saturation related effect at the 1% level, beams with a lower intensity experienced a greater increase in η than beams with a higher intensity, however the quantum efficiency may not be uniform over the diode – near the edges η is greater. The difference between the DC and AC response could only be described by a non-

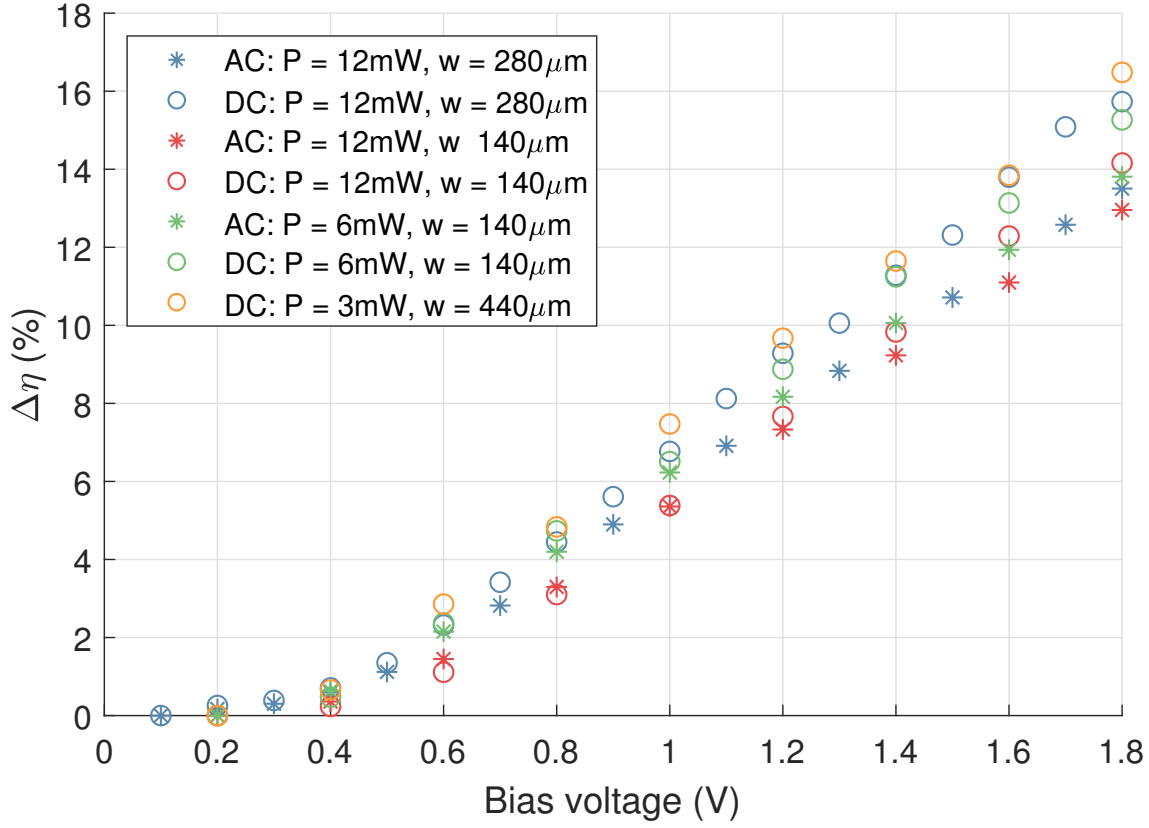


Figure 1.3

linear process. These effects are small, and Figure 1.3 gives a clear incentive to increase the bias voltage.

Figure 1.4 shows the photocurrent measured has the noise level predicted by Equation 1.1, meaning that the increase in photocurrent at a reverse bias voltage of 0.6V is, without doubt, an increase in quantum efficiency. This measurement was performed with 4 different photocurrents, ranging from 7.5 mA to 15.5 mA, and over this range there was a 2% change in the ratio between the DC and AC measurement, this is shown in Figure 1.5. The SNR of this measurement is limited to the 1% level due to the difference between the calibration peak applied at 1111 Hz and the level

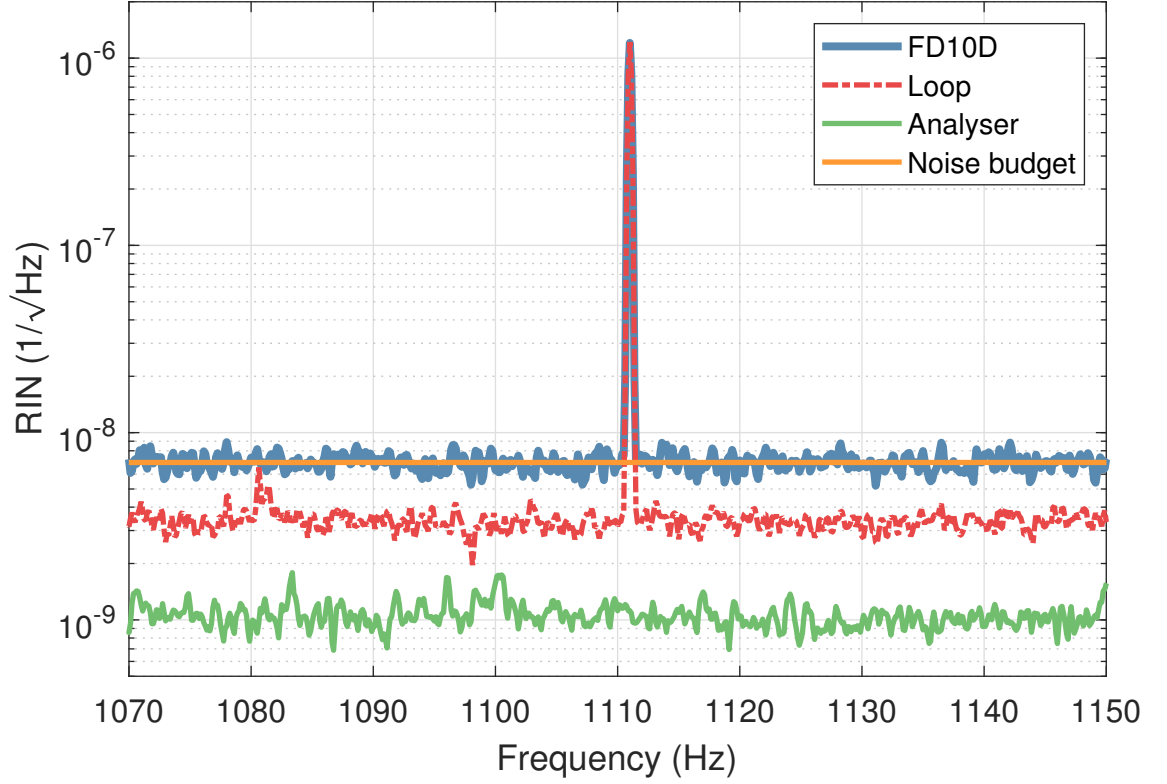


Figure 1.4

of shot noise.

1.5.4 Saturation Limit for some laser components photodiodes

The laser components photodiodes were exposed to a light signal of $\text{RIN} = 12\text{e-}6$. Their bias was set to the maximum that was compatible with $1/f$ noise being below shot noise (see Section ??). The power of the light was increased, and the decrease in the relative intensity of the AC modulation was recorded. This is shown in Figure 1.6. A decrease in AC signal indicates that the diode is saturating. These diodes cannot handle the power levels required by for use in an GW interferometer. This

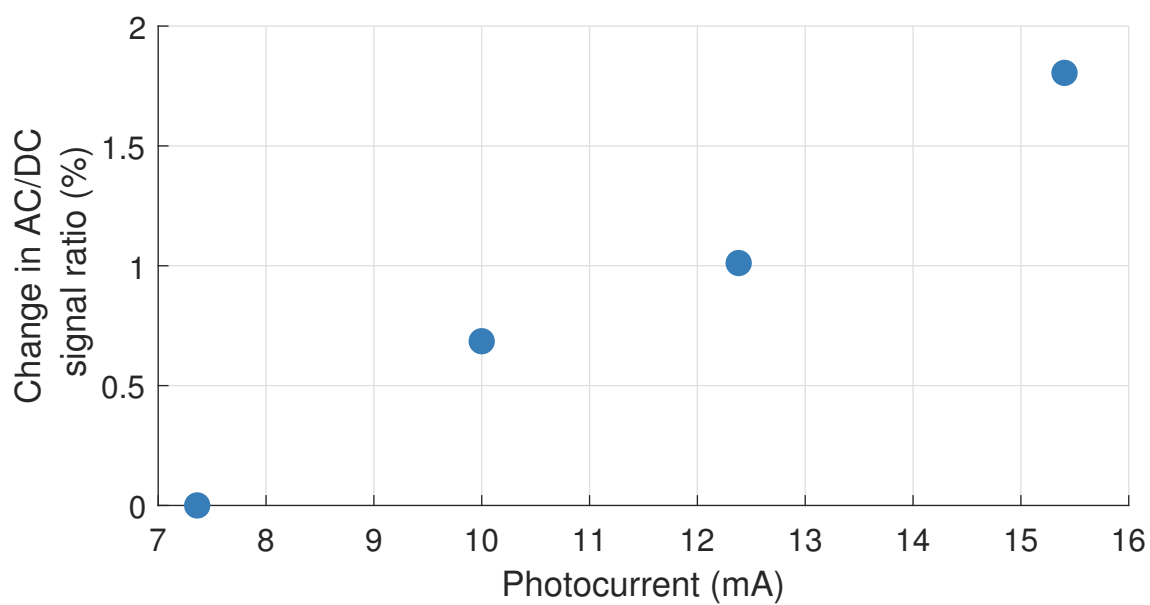


Figure 1.5

problem was not evident in the FD10D (see figure 1.5).

1.6 Dark Current and Dark Noise in Extended In-GaAs Photodiodes

formulas etc.

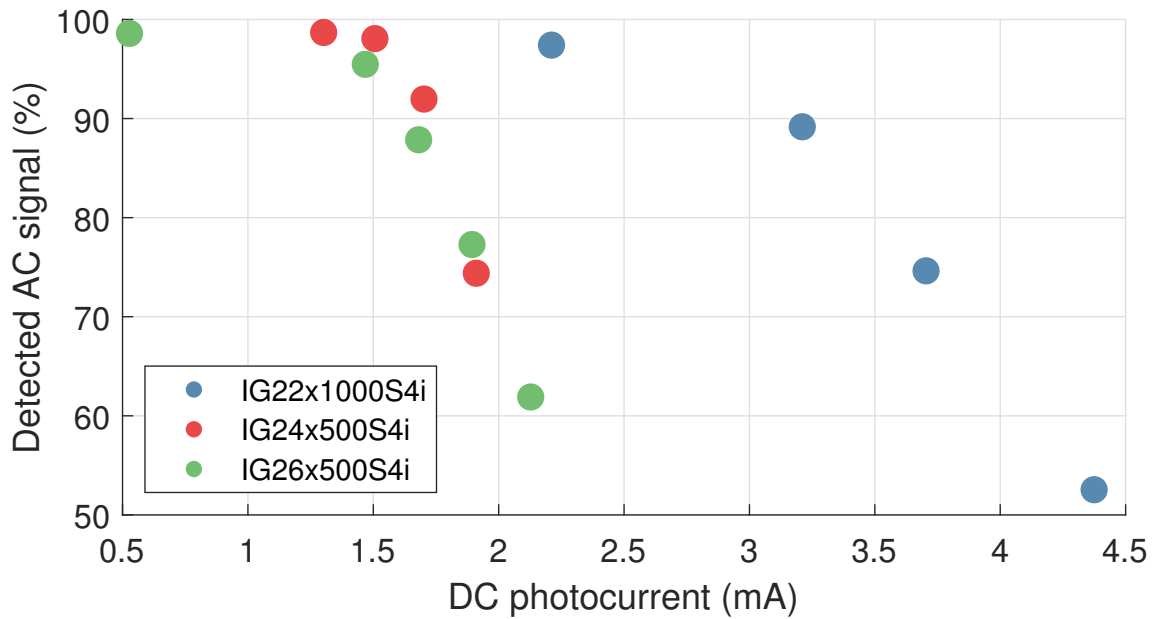


Figure 1.6

1.7 The Relationship Between a Diode's Bias Voltage, Quantum Efficiency and Linearity

1.8 Discussion

1.9 Conclusion

1.10 Notes about the Diodes Tested

This is a candid (for now) set of notes about each diode to help me know what evidence I actually have. I only measured the dependence of quantum efficiency on reverse bias with one photodiode – FD10D. If a diode was shown to be non-linear for a low photocurrent then testing with it stopped.

1.10.1 FD10D

peak QE wavelength = $\sim 2\mu\text{m}$, cut off wavelength = $2.6\mu\text{m}$

This dark current has been measured as a function of temperature and as a function of bias. The dark noise has been measured as a function of temperature and bias. It was found that 0.6V is a ‘suitable’ bias voltage to use at room temperature.

With a 0.6V reverse bias, the diode was linear up to 16mA. This wasn’t measured until it became non-linear. The spot size could have been made bigger, but 16mA is a good amount of photocurrent.

The quantum efficiency as a function of bias was measured. This was calibrated against a power meter at 0.6V reverse bias. It was found that the QE was 73%. The photocurrent was shown to be shot noise limited, and so we can say this was a real increase in shot noise.

The impedance of the photodiode was measured. It depended on the light level, and exhibited a ‘negative capacitance’ effect – although this is not really well understood.

1.10.2 IG19x1000S4i

As this diode has a cut-off slightly out of the range of interest, not a lot of experiments were done with it. This diode has a cut-off at $1.9\mu\text{m}$. The QE is about 80% from about 1.2 to $1.7\mu\text{m}$. This dark current has been measured as a function of

bias. The dark noise has been measured as a function of bias. It was found that that measurements up to 20mA of photocurrent could be made without saturating. The reverse bias could be turned all the way up to 1V.

1.10.3 IG22x1000S4i

peak QE wavelength = $\sim 1.8\mu\text{m}$, cut off wavelength = $2.15\mu\text{m}$

The dark noise and dark current were measured for this diode at room temperature. In order to keep 1/f noise low, the reverse bias voltage must be less than 0.06V. This is probably due to the size of the diode. The diode needs to have a bias greater than this to be linear. For this reason, no more measurements were made with it. Additionally, this diode appears to have broken at some point and so further experiments with it cannot be made.

1.10.4 IG24x500S4i

peak QE wavelength = $\sim 2\mu\text{m}$, cut off wavelength = $2.35\mu\text{m}$

This dark current has been measured as a function of temperature and as a function of bias. The dark noise has been measured as a function of temperature and bias. It was found that 0.275V is a 'suitable' bias voltage to use at room temperature. Cooling it down from 30°C to -40°C reduced the noise by a factor of 3.

The forward voltage characteristics of this diode were used to see what the ideality factor of the diode was. This was found to be close to 1 (i.e. it is ideal).

This diode was found to be produce a excess white noise even with low amounts of power on it. It is likely that this was due to saturation. Regardless, this diode was found to be unsuitable for making shot noise limited measurements.

1.10.5 IG26x500S4i

peak QE wavelength = $\sim 2\mu\text{m}$, cut off wavelength = $2.45\mu\text{m}$ This dark current has been measured as a function of temperature and as a function of bias. The dark noise has been measured as a function of temperature and bias. It was found that 0.167V is a 'suitable' bias voltage to use at room temperature. Cooling it down from 20°C to -40°C reduced the noise by a factor of 3.

This diode was only linear up to a very low amount of photocurrent ($<1\text{mA}$). Further tests on this diode were vetoed because of this.

1.10.6 G12182-010k

peak QE wavelength = $\sim 1.7\mu\text{m}$, cut off wavelength = $2.1\mu\text{m}$

This dark current has been measured as a function of temperature and as a function of bias. The dark noise has been measured as a function of temperature and bias. It was usable at all bias voltages.

1.10.7 G12182-005k

peak QE wavelength = $\sim 1.7\mu\text{m}$, cut off wavelength = $2.1\mu\text{m}$

This dark current has been measured as a function of temperature and as a function of bias. The dark noise has been measured as a function of temperature and bias. It was usable at all bias voltages.

1.10.8 G12183-010k

peak QE wavelength = $\sim 2\mu\text{m}$, cut off wavelength = $2.6\mu\text{m}$

This dark current has been measured as a function of temperature and as a function of bias. The dark noise has been measured as a function of temperature and bias. It was usable up to 0.1V bias voltage at room temperature. Cooling it to -40°C made it usable up to 1V reverse bias.

1.10.9 G12183-005k

peak QE wavelength = $\sim 2\mu\text{m}$, cut off wavelength = $2.6\mu\text{m}$

This dark current has been measured as a function of temperature and as a function of bias. The dark noise has been measured as a function of temperature and bias. It was usable up to 0.6V bias voltage at room temperature. Cooling it to -30°C made it usable up to 1V reverse bias.

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